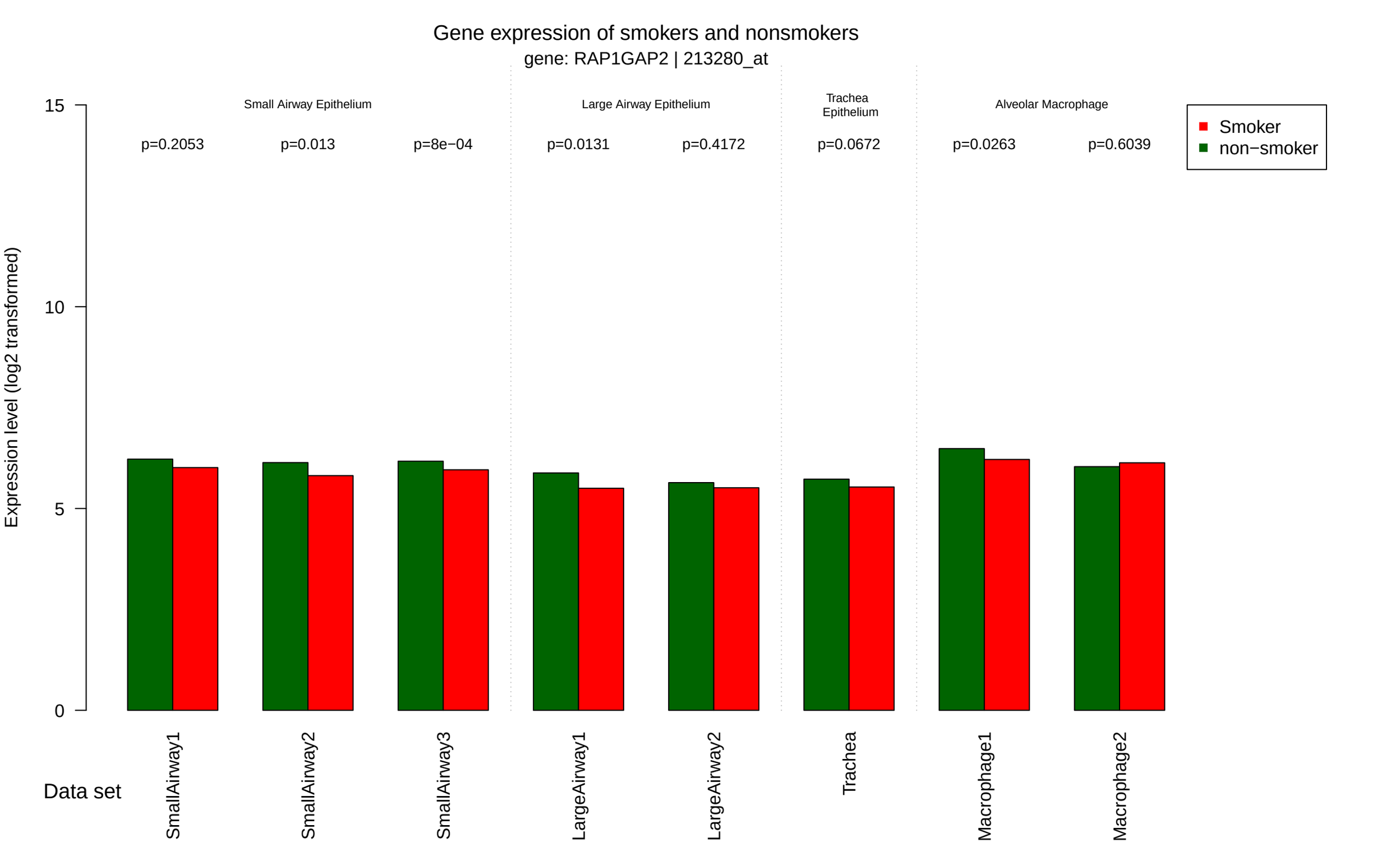
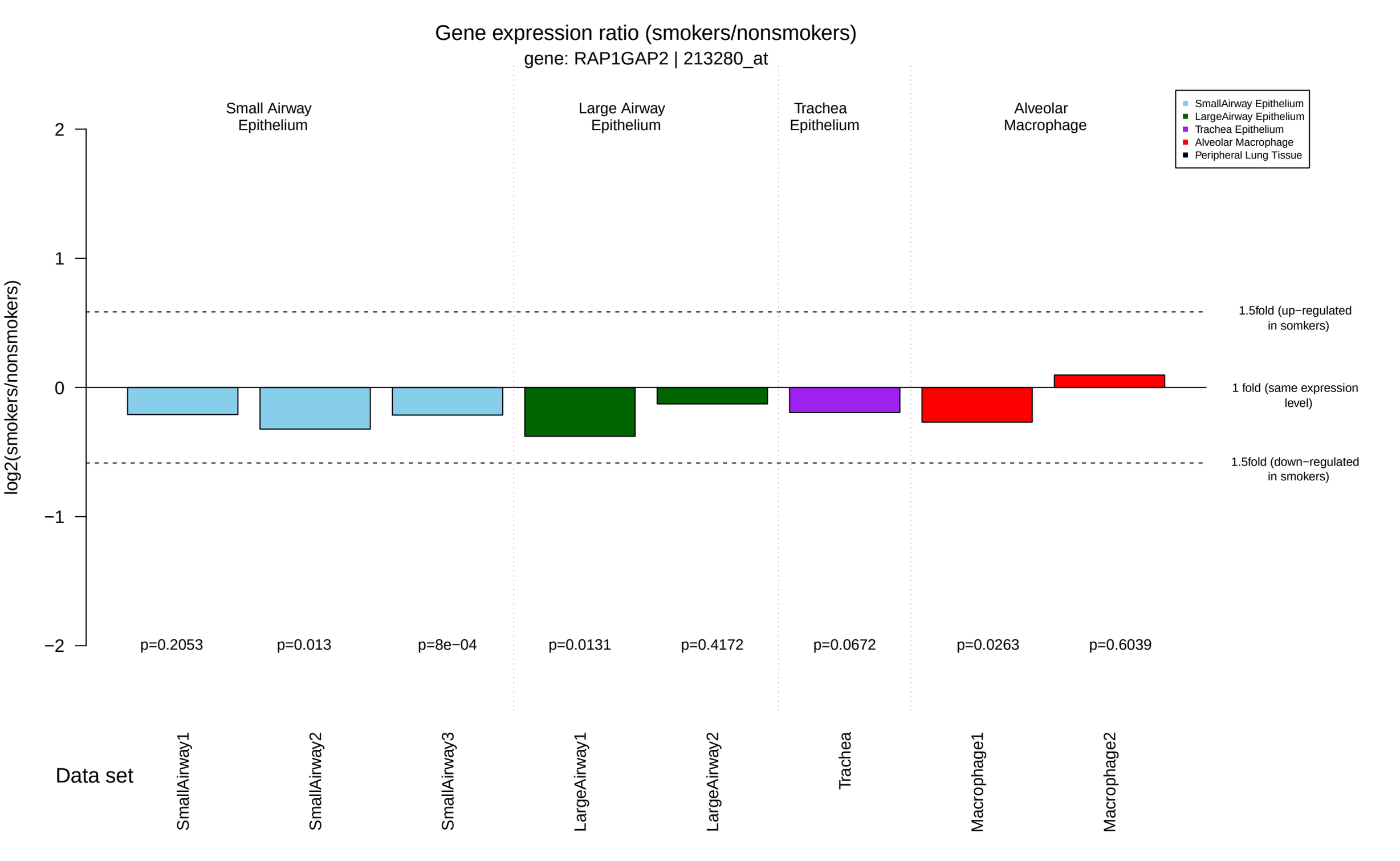
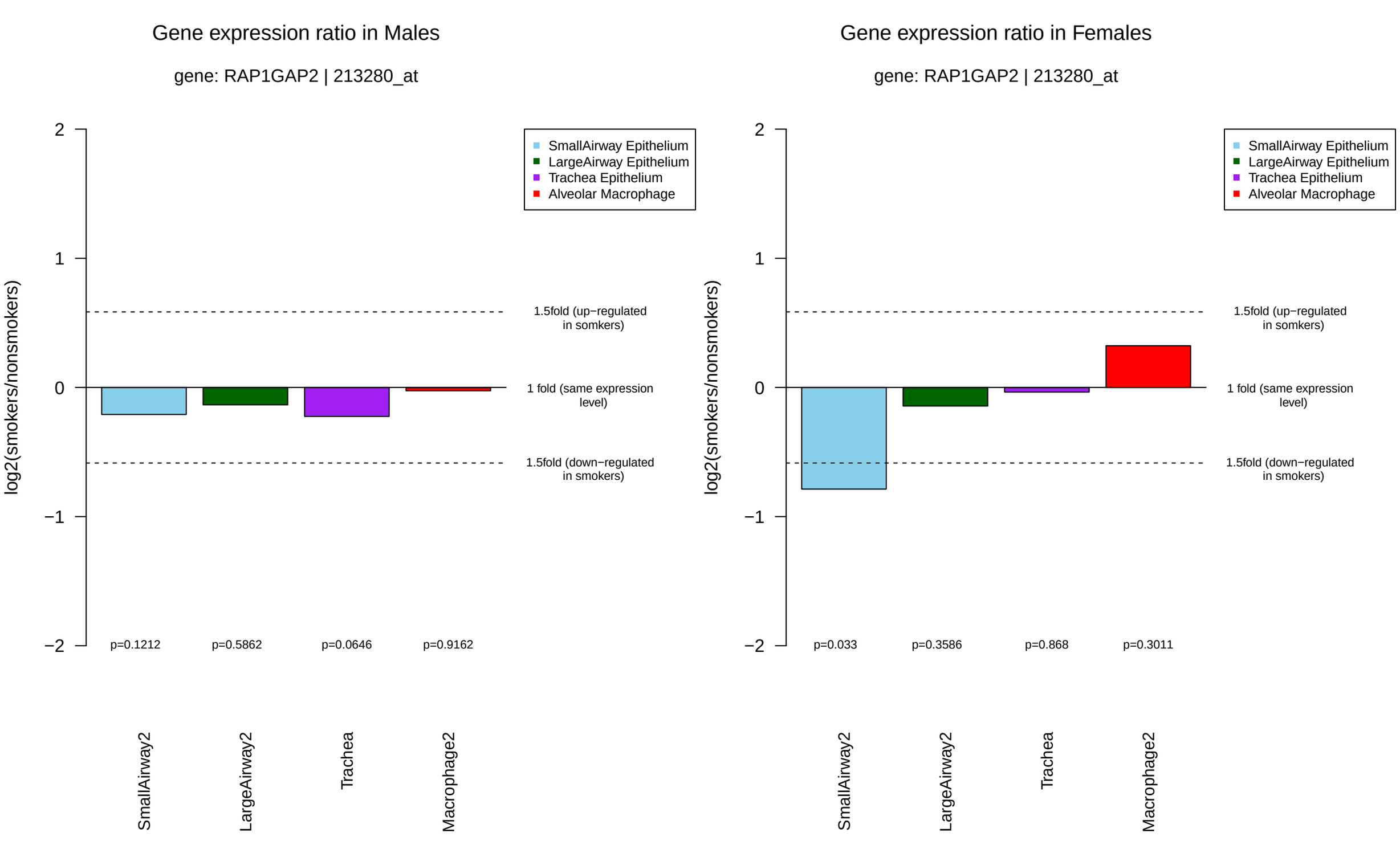
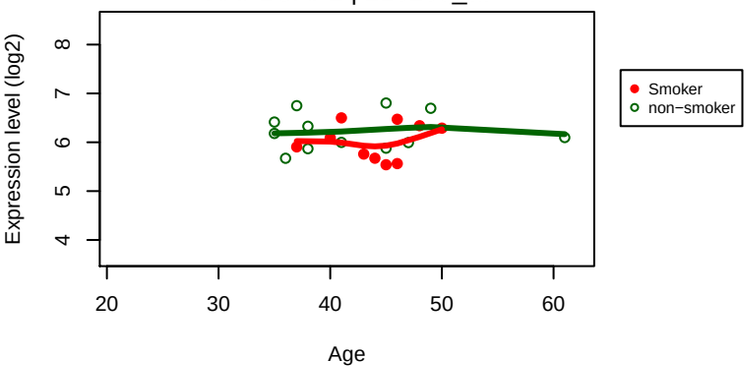
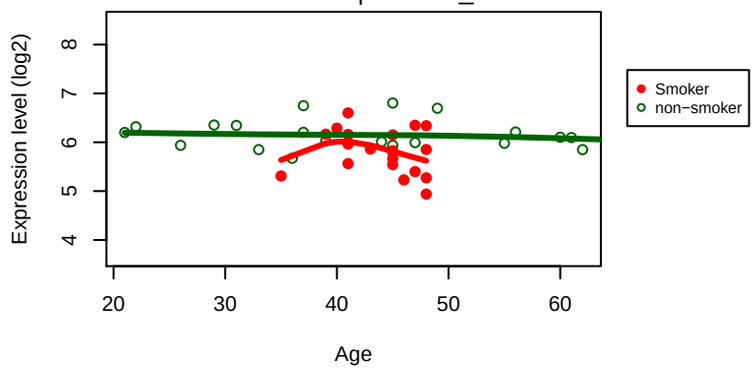
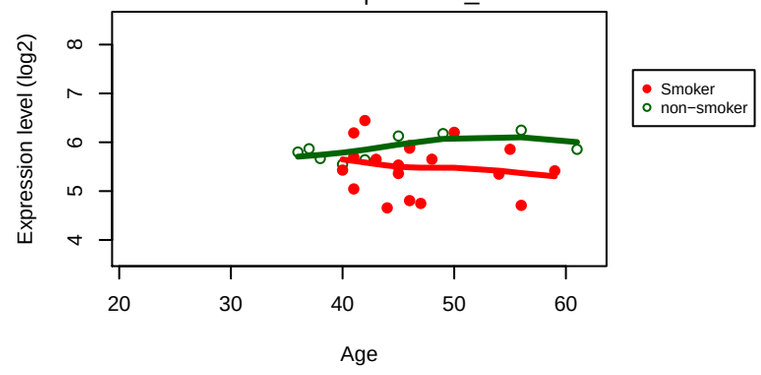

Gene expression of smokers and nonsmokers
gene: RAP1GAP2 | 213280_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2053
p=0.013
p=8e−04
p=0.0131
p=0.4172
p=0.0672
p=0.0263
p=0.6039
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

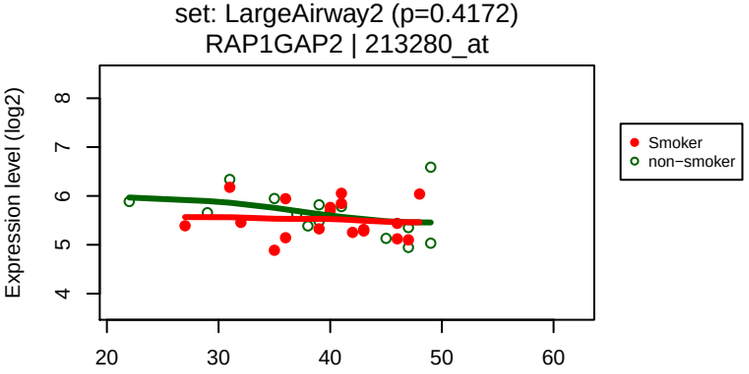

Gene expression ratio (smokers/nonsmokers)
gene: RAP1GAP2 | 213280_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2053
p=0.013
p=8e−04
p=0.0131
p=0.4172
p=0.0672
p=0.0263
p=0.6039
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

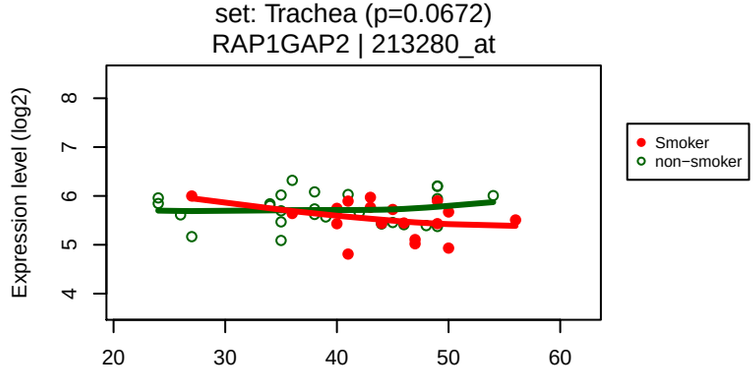

Gene expression ratio in Males
gene: RAP1GAP2 | 213280_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1212
p=0.5862
p=0.0646
p=0.9162
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAP1GAP2 | 213280_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.033
p=0.3586
p=0.868
p=0.3011
SmallAirway2
LargeAirway2
Trachea
Macrophage2

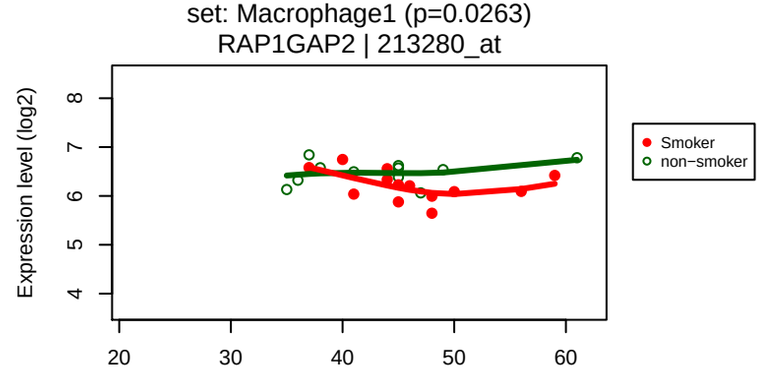

set: SmallAirway1 (p=0.2053)
RAP1GAP2 | 213280_at
Expression level (log2)
Age
Smoker
non−smoker

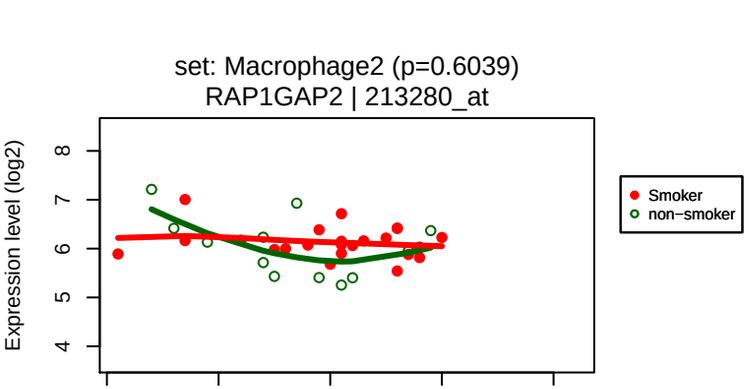

set: SmallAirway2 (p=0.013)
RAP1GAP2 | 213280_at
Expression level (log2)
Age
Smoker
non−smoker

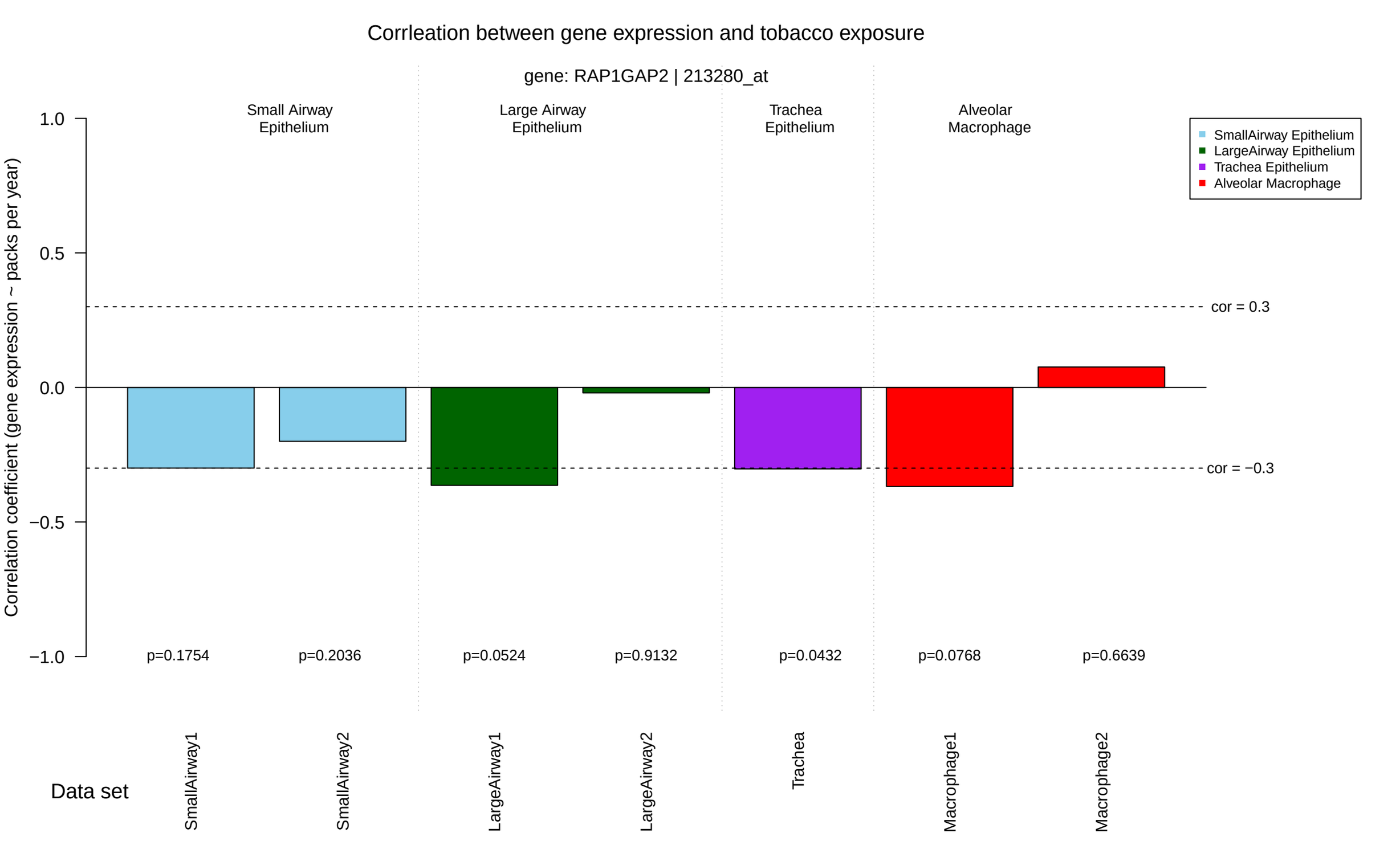

set: LargeAirway1 (p=0.0131)
RAP1GAP2 | 213280_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4172)
RAP1GAP2 | 213280_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0672)
RAP1GAP2 | 213280_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0263)
RAP1GAP2 | 213280_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6039)
RAP1GAP2 | 213280_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAP1GAP2 | 213280_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1754
p=0.2036
p=0.0524
p=0.9132
p=0.0432
p=0.0768
p=0.6639
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2